NetApp Knowledge Base

How to update existing KMIP SSL certificates from ONTAP boot menu

https://kb.netapp.com/on-prem/ontap/DM/Encryption/Encryption-KBs/How_to_update_existing_KMIP_…
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Applies to

- ONTAP 9.8 and later
- External Key Manager (EKM)

Description

Certain scenarios require KMIP SSL certificate update for external key management server to allow ONTAP to boot

NOTE: The external key management server is a third-party system in your storage environment that serves authentication keys to nodes using the Key Management Interoperability Protocol (KMIP). The cluster and KMIP server use KMIP SSL certificates to verify each other's identity and establish an SSL connection.

Procedure

Preparation

The following detail will be necessary for KMIP certificate renewal:

1. Confirm the cluster is using external key manager.
2. You must have obtained the replacement public certificate for the KMIP server (KMIP SSL server-ca certificate).
3. You must have obtained the replacement public certificate and private key for the cluster (KMIP client certificate)

Process

1. From the ONTAP boot menu enter systemshell to display the existing certificates installed then remove the existing certificate.

```
[...]

(11) Configure node for external key management.
Selection (1-11)? systemshell

Forking /bin/sh
pid: 3595

# cat /cfcard/kmip/certs/client.crt
# cat /cfcard/kmip/certs/client.key
# cat /cfcard/kmip/certs/CA.pem
#
# rm -f /cfcard/kmip/certs/client.crt
# rm -f /cfcard/kmip/certs/client.key
# rm -f /cfcard/kmip/certs/CA.pem
# exit
```

2. From the ONTAP boot menu select option **11**.

```
[...]

(11) Configure node for external key management.
Selection (1-11)? 11
```

3. When prompted confirm you have gathered the required information from the Preparation section of this article.

```
Do you have a copy of the /cfcard/kmip/certs/client.crt file? {y/n} y

Do you have a copy of the /cfcard/kmip/certs/client.key file? {y/n} y

Do you have a copy of the /cfcard/kmip/certs/CA.pem file? {y/n} y

Do you have a copy of the /cfcard/kmip/servers.cfg file? {y/n} y
```

4. Supply the information for each of the prompts:

```
Enter the client certificate (client.crt) file contents:
Enter the client key (client.key) file contents:
Enter the KMIP server CA(s) (CA.pem) file contents:
Enter the server configuration (servers.cfg) file contents:
```

Example

```
Enter the client certificate (client.crt) file contents:
-----BEGIN CERTIFICATE-----
MIIDvjCCAqagAwIBAgICN3gwDQYJKoZIhvcNAQELBQAwgY8xCzAJBgNVBAYTAlVT
MRMwEQYDVQQIEwpDYWxpZm9ybmlhMQwwCgYDVQQHEwNTVkwxDzANBgNVBAoTBk5l
MSUbQusvzAFs8G3P54GG32iIRvaCFnj2gQpCxciLJ0qB2foiBGx5XVQ/Mtk+rlap
Pk4ECW/wqSOUXDYtJs1+RB+w0+SHx8mzxpbz3mXF/X/1PC3YOzVNCq5eieek62si
Fp8=
-----END CERTIFICATE-----

Enter the client key (client.key) file contents:
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEAoU1eajEG6QC2h2Zih0jEaGVtQUexNeoCFwKPoMSePmjDNtrU
MSB1SlX3VgCuElHk57XPdq6xSbYlbkIb4bAgLztHEmUDOkGmXYAkblQ=
-----END RSA PRIVATE KEY-----

Enter the KMIP server CA(s) (CA.pem) file contents:
-----BEGIN CERTIFICATE-----
MIIEizCCA3OgAwIBAgIBADANBgkqhkiG9w0BAQsFADCBjzELMAkGA1UEBhMCVVMx
7yaumMQETNrpMfP+nQMd34y4AmseWYGM6qG0z37BRnYU0Wf2qDL61cQ3/jkm7Y94
EQBKG1NY8dVyjphmYZv+
-----END CERTIFICATE-----
```

```
Enter the IP address for the KMIP server: 10.10.10.10
Enter the port for the KMIP server [5696]:

System is ready to utilize external key manager(s).
Trying to recover keys from key servers....
kmip_init: configuring ports
Running command '/sbin/ifconfig e0M'
..
..
kmip_init: cmd: ReleaseExtraBSDPort e0M
```

5. The recovery process will complete:

```
System is ready to utilize external key manager(s).
Trying to recover keys from key servers....
[Aug 29 21:06:28]: 0x808806100: 0: DEBUG: kmip2::main: [initOpenssl]:460:
Performing initialization of OpenSSL
Successfully recovered keymanager secrets.
```

6. Select option **1** from the boot menu to continue booting into ONTAP.

```
*****************************************************************************
* Select option "(1) Normal Boot." to complete the recovery process.
*
*****************************************************************************

(1)  Normal Boot.
[...]
Selection (1-11)? 1

 System is ready to utilize external key manager(s).
```

7. Once booted, need to follow the certificate installation procedure using CLI/REST to update the certificates.

Additional Information

- [Replace KMIP SSL certificates on ONTAP cluster]()

- Feature request: Provide key manager (external and onboard) recovery options for the system admin in the boot menu